

A Low-Cost Single-Camera Adaptive Signal Controller for Heterogeneous Urban Traffic: Self-Calibrating Pressure Control with Bounded Fairness and Emergency Preemption

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Abstract

Fixed-time signal plans remain the default in many South Asian cities, where traffic is motorcycle-dominated, demand is strongly asymmetric between approaches, and the budget for inductive loops or radar is unavailable. We present R.E.L.A.Y., an adaptive signal controller whose only input is one existing fixed junction camera. A single convolutional detector produces per-approach, per-class vehicle counts, which are converted to passenger-car-unit (PCU) demand and consumed by a queue-weighted pressure controller in the max-pressure family. The controller adds the operational rules a deployable signal needs and a plain argmax policy lacks: empty-phase skipping, gap-out, minimum and maximum green, a hard anti-starvation bound, pedestrian demand treated as pressure rather than as a fixed interval, and emergency preemption with a bounded hold. It also estimates each approach’s arrival and discharge rate from the same counts, with no site survey and no configured saturation flow, and uses those estimates for two decisions a constant cannot make: the green is sized from Webster’s cycle evaluated on the measured degree of saturation, and a phase change is priced at the discharge its clearance interval forfeits. We evaluate it in a seeded point-queue microsimulation against an equal-split fixed-time plan and against a deliberately favourable reference whose cycle and splits come from Webster’s formulae applied to the true mean demand. Across 30 seeds per cell, three demand shapes and five demand levels, mean delay falls by 28.4% to 44.3% against the Webster plan and by 27.9% to 96.1% against the equal-split plan, with the 95th-percentile wait cut from 25.8 s to 14.5 s at 1278 veh/h. A wider robustness sweep of 72 cells, spanning seven demand shapes, two junction topologies and loads to 2462 veh/h, beats both baselines in 71; the exception is a single approach demanding 82% of all green, where no cycle short enough to respect the configured fairness bound is stable. We report one negative result: in a three-junction corridor, disabling the downstream coupling term changes mean delay by at most 0.9%, so the corridor gain of 44.6% to 57.9% comes from local adaptivity rather than from coordination. Emergency preemption reaches green in 4.09 s on average over 900 trials, with a measured worst case of 13.99 s against an analytical bound of 14.0 s. The perception stack runs at a median of 38.3 ms per frame at 960 px inference on a consumer laptop, but on motorcycle-dense footage it undercounts severely, with sampled counts rising 83% between 640 px and 1600 px without converging. This is a simulation and real-footage feasibility study, not a field deployment.

Keywords: adaptive traffic signal control; max-pressure control; computer vision; heterogeneous traffic; emergency vehicle preemption; low-cost intelligent transport systems

1 Introduction

Urban signal control in low- and middle-income cities is usually performed by fixed-time plans or by police officers operating signals manually. Both allocate green without measuring what is

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queued. The consequence is visible at almost any asymmetric junction: a busy arterial approach holds a long queue while the crossing minor street receives the green it would receive at three in the morning. Webster’s method [1] showed that cycle length and splits should follow demand, but a fixed plan can only follow the demand measured when the plan was designed, and re-timing surveys are infrequent. Studies of signalised intersections in the Kathmandu valley report saturation flows and delays that differ substantially from standard capacity guidance, attributing the difference to the local vehicle mixture and to lane discipline [2, 3], which is exactly the condition under which a plan imported from a design manual performs badly.

The established remedy is a demand-responsive system such as SCOOT [4, 5] or SCATS [6], or an optimisation-based controller in the OPAC [7] and RHODES [8] lineage. These work and have decades of field evidence behind them, but they are expensive: they assume a detector plant, communications and a central control room, so per-junction cost is dominated by sensing and civil works rather than by the algorithm. In a city where the signal head itself is sometimes unpowered, that capital profile is prohibitive.

Two conditions have changed. Fixed cameras are already installed at many urban junctions for enforcement and monitoring, so a camera-based controller reuses a paid-for asset. And single-stage object detectors are now accurate and cheap enough to run at video rate on commodity hardware [9, 10], which makes per-approach vehicle counting a software problem rather than a sensor-procurement problem. Vision-based queue and density estimation from fixed cameras is an established technique in its own right [11–13], and several recent studies connect a detector to a signal timer on this basis [14–17].

What is often missing in that applied literature is the control side. Many camera-based studies compute a vehicle count and set green time in proportion to it, which is a proportional rule rather than a control policy, and they rarely state what prevents a minor approach from being starved, what happens when the detector fails, or how the policy behaves at saturation. Conversely, the theoretical literature on max-pressure control [18, 19] provides throughput-optimality results in a form that assumes ideal queue observation and imposes no minimum green, no clearance interval and no fairness bound, all of which a deployable signal requires.

This paper describes and evaluates R.E.L.A.Y., a controller placed deliberately between those positions. Its contributions are:

- A single-camera perception-to-control pipeline for heterogeneous traffic in which presence and PCU-weighted demand are kept as separate quantities, because a PCU-weighted score alone classifies a lone motorcycle as an empty lane.
- A self-calibrating pressure controller. Arrival and discharge rates are estimated per approach from the count sequence alone (Section 4.3) and used for two decisions that a constant cannot make: each stage’s green is floored at its share of Webster’s cycle evaluated on the measured degree of saturation (Section 4.4), and a stage change must beat the discharge its clearance interval forfeits (Section 4.5). No saturation-flow table, traffic survey or per-site constant is required. We show these are not refinements: without them an unconstrained pressure rule terminates every green at the safety minimum once demand approaches capacity, and loses to a fixed plan outright.
- Fairness enforced by explicit bounds rather than by the score: separate maximum-wait bounds for vehicles and pedestrians, a cap on the preemption hold, and an anti-starvation rule that serves the worst-treated queue instead of rotating among equally starved ones, which is what keeps the rule meaningful beyond capacity.
- An evaluation against a Webster plan computed from the true mean demand, much harder to beat than the equal-split baseline camera-based studies usually use, reported per cell over 72 cells of demand shape, topology and load rather than as an average, together with a measured preemption latency distribution compared against a closed-form worst-case bound.

We are explicit about what this is not. There is no field installation, no controller cabinet in the loop and no before-and-after study on a real signal: control results come from a seeded microsimulation and perception results from recorded footage. The contribution is a feasibility and design study at the level a city could use to decide whether a pilot is worth funding, and Section 7 states which claims a field trial would still have to establish.

2 Related work

2.1 Fixed-time and demand-responsive control

Webster’s analysis of cycle length and green splits [1] remains the reference method for fixed-time plans and is the basis of the stronger baseline used here. Among demand-responsive systems, SCOOT adjusts splits, offsets and cycle time from cyclic flow profiles measured upstream of the stop line, SCATS selects among plans using measured degree of saturation, and OPAC and RHODES solve a rolling-horizon optimisation over predicted arrivals. All four depend on a detector plant installed and maintained in the carriageway, and all four were designed for streams in which lane discipline holds and vehicle classes are similar in size.

2.2 Max-pressure and backpressure control

Max-pressure signal control derives from backpressure scheduling in queueing networks [20]. Varaiya [18, 19] showed that repeatedly activating the phase maximising a weighted difference between upstream and downstream queues stabilises any demand within the capacity region using only local queue information, with no demand forecast and no coordination plan, which is attractive for a decentralised low-communication deployment. Related work established distributed formulations [21] and decentralised variants with weaker information requirements [22].

Two properties of the theoretical policy conflict with practice. First, the argmax is evaluated freely at every decision epoch, so the policy may switch as often as the decision interval allows, and since each switch costs a clearance interval during which no approach discharges, unconstrained switching wastes capacity; Levin et al. [23] address this with a cyclical phase structure, and our balanced-demand result in Section 6 is an independent, quantified instance of the same effect. Second, throughput optimality is proved for the unconstrained policy, and adding minimum green, maximum green, hysteresis and forced service breaks the argmax property and voids the proof. We therefore describe our controller as a queue-weighted pressure heuristic in the max-pressure family and make no stability claim.

2.3 Learning-based control

Reinforcement learning is the dominant research direction for signal control. Surveys by Wei et al. [24], Haydari and Yilmaz [25] and Noeen et al. [26] cover the field; representative works include IntelliLight [27], large-scale decentralised training [28] and PressLight [29], whose reward is derived from max-pressure and which is therefore a direct point of comparison. Early work by Genders and Razavi [30] established the basic formulation.

We did not adopt a learned policy, for reasons specific to the deployment context rather than to the merits of the methods: training requires a simulator calibrated to the target junction and none exists for the junctions we are concerned with; a neural policy does not naturally expose the maximum-wait or clearance guarantees an authority approving a signal plan needs; and it transfers poorly across geometries, whereas a pressure rule parameterised only by a phase set moves to a new junction by drawing polygons on a camera image. The comparison we do not make, and which should be made, is a head-to-head against a trained policy on the same arrival streams.

2.4 Camera-based counting and vision-driven signal control

Vehicle detection from fixed cameras is a mature application of single-stage detectors [9, 10], reviewed by Terven et al. [31], typically trained on general corpora [32] and extended with tracking when trajectories are required [33, 34]. Our system uses the Ultralytics implementation of the YOLO family [35]; the architecture generation we fine-tune is described in [36], which remains a preprint. Because the controller consumes instantaneous per-approach occupancy rather than cumulative crossings, we count detections inside approach polygons instead of tracking individual vehicles, which removes identity-switch errors in dense queues from the control path.

Recent applied studies pair such a detector with a signal timer. Papanashi et al. [14] report YOLO-based adaptive timing evaluated in Bengaluru, a heterogeneous-traffic context comparable to ours; Alruyshid et al. [15] add emergency and public-bus prioritisation in a Baghdad case study; Talaat et al. [16] and Medvei et al. [17] report detector-driven controllers, the latter combining detection with a learned policy. These works establish that the perception component is practical. Our contribution relative to them is on the control side: an explicit pressure formulation, stated fairness bounds, a preemption latency bound, and an evaluation including a demand-optimal fixed-time baseline that reports where the adaptive policy loses. We are not aware of a published camera-based adaptive controller that reports a regime in which its own policy is worse than the fixed-time baseline; given the mechanism identified in Section 6.2 that regime almost certainly exists for those systems too, and its absence from the record is more likely a reporting bias than a property of the systems.

Emergency preemption has its own long operational history and a careful literature on its side effects, since a preemption interrupts the cycle and can create hazards at the transition [37]; Humagain et al. [38] review route optimisation and preemption methods, and recent work exploits connected-vehicle messaging to grant priority with explicit dilemma-zone protection [39]. Our approach differs in its detection channel, since the emergency vehicle is recognised visually by the same detector that counts ordinary traffic and no transponder or vehicle-side equipment is required. That is an advantage in cost and a disadvantage in reliability, which is why the ambulance class carries the highest confidence threshold and why the hold is capped, and what we add is a stated worst-case latency bound with a measurement against it.

3 System architecture

R.E.L.A.Y. consists of a perception stage and a control stage connected by a single data structure: a mapping from approach identifier to per-class vehicle count. Everything downstream of that mapping is independent of how the counts were obtained, which is what allows the same controller to be driven by recorded footage, a live camera, or a synthetic feed.

3.1 Perception

A single fixed camera views the junction. Each approach is defined by a polygon in normalised image coordinates, drawn once per site; no per-site model training is performed. For each frame a YOLO detector produces class-labelled boxes, which are filtered in four steps.

First, per-class confidence gating, because a single global threshold is unsuitable for mixed traffic: two-wheelers are small in the image and are lost at a threshold that suppresses false positives on cars, so the system uses 0.24 for motorcycles and bicycles, 0.38 by default, 0.50 for trucks and 0.68 for ambulances. The high ambulance gate is a safety choice, not an accuracy choice, since an ambulance detection moves the signal by itself and a marginal read on a white van would hand the junction to a non-existent emergency, while a vehicle below the gate still contributes ordinary demand through its next-best label. Second, cross-class deduplication, since per-class non-maximum suppression can leave one distant vehicle carrying stacked car, truck and motorcycle

boxes: a greedy pass in descending confidence retains a box only if its intersection over union with every retained box is below 0.5, running after the confidence filters so a discarded low-confidence box cannot suppress a valid one. Third, approach assignment by centroid containment, giving one consistent rule for vehicles straddling a boundary, followed by an exponential moving average per approach and class with coefficient 0.35, roughly a one-second time constant at the observed detection rate. Fourth, failure handling: a bad frame returns the last smoothed counts rather than raising, and a staleness flag set after three seconds without a successful update makes the pipeline feed the controller empty counts, so the wait clocks and the maximum-wait rule cycle the phases safely instead of holding green on stale demand.

3.2 Control, topology and deployment

The controller consumes the count mapping and a set of conflict-free phases, and is topology-agnostic: four-arm, three-arm and two-arm junctions differ only in the phase set passed at construction, and approaches may be specified per arm or per movement, in which case the phase set is generated to include paired through phases, paired protected-left phases and one split phase per arm serving both of that arm’s movements. Adding a junction therefore requires drawing polygons and declaring phases, not changing code.

Pedestrian demand enters as a separate transponder-style input giving, per arm, the number waiting, the longest wait and the number currently crossing. It is sanitised before use: non-numeric values, unknown arms, negative counts and non-finite waits are discarded, counts are clamped to 200 and waits to 180 s, so a client fault cannot inject an unbounded aging term.

The intended installation is one small computer per junction, taking a feed from an existing camera and driving the existing signal controller, with no central server in the control path and no junction-to-junction communication beyond the single downstream queue scalar the corridor configuration reads. We measure the perception cost in Section 6 but claim no cost figure, since no hardware was procured.

4 Control formulation

4.1 Notation and derived quantities

Let A be the set of approaches and P a set of phases, each $p \in P$ a subset of A that may be served simultaneously without conflict. Let Δt be the control step. At each step perception reports counts $n_{a,c}$ for approach a and class c , and a set of approaches on which an emergency vehicle was detected in the current frame.

Two distinct quantities are derived, the PCU-weighted demand and the occupancy indicator:

$$d_a = \sum_c w_c n_{a,c}, \quad (1)$$

with weights w_c from Table 1, and

$$o_a = \mathbb{1}[\sum_c n_{a,c} > 0]. \quad (2)$$

Keeping these separate matters: a single motorcycle contributes $d_a = 0.3$, which a demand threshold would treat as an empty approach, so the empty-phase skip and gap-out rules key on o_a rather than d_a and a lone two-wheeler still receives a full minimum green.

Each approach carries a clock r_a answering how long the oldest waiting vehicle on that approach has waited, rather than how long since the approach last had green. With G the set of approaches

Table 1: Passenger-car-unit weights w_c used in Equation (1). These are configuration values selected for a motorcycle-dominated mix. Their ordering is consistent with published equivalency estimates for heterogeneous South Asian traffic [40–42], but they were not estimated from local field measurements in this work; see Section 7.

Class	car	motorcycle	bus	truck	ambulance	bicycle	auto-rickshaw
w_c	1.0	0.3	2.5	2.5	2.0	0.2	0.8

currently green and z_a counting consecutive seconds the approach has read empty,

$$r_a \leftarrow \begin{cases} 0 & a \in G, \\ r_a + \Delta t & a \notin G \text{ and } o_a = 1, \\ 0 & a \notin G, o_a = 0, z_a \geq \tau_e, \\ r_a & \text{otherwise,} \end{cases} \quad (3)$$

with $\tau_e = 3$ s. The third case is necessary rather than cosmetic: without it a long-vacant approach accumulates a large clock value, which the first arriving vehicle inherits, triggering forced service against a loaded corridor. The debounce keeps a detection dropout from erasing a real queue’s guarantee.

4.2 Phase score

Define a bounded urgency multiplier $u_a = 1 + \min(r_a, R)/R \in [1, 2]$, where R is the maximum-wait parameter. The vehicle pressure of a phase is

$$D(p) = \sum_{a \in p} d_a u_a. \quad (4)$$

Aging multiplies each approach’s own demand and is capped at a factor of two, so waiting cannot manufacture demand that does not exist. This is a deliberate departure from an additive term $d_a + \lambda r_a$, which inverts the intended ordering: with $\lambda = 0.4$ a two-vehicle approach that has waited 45 s scores like an eighteen-PCU queue and the controller rotates phases at minimum-green cadence. For the same reason the multiplier is applied per approach rather than at phase level, since a phase-level $\max_a r_a$ lets a starving single-vehicle turn bay double the score of a phase whose large through queue has just been served.

Pedestrians are treated as demand. A phase p frees the crossing of arm x when neither x nor the opposing arm, whose straight-ahead movement would traverse the same crossing, is served: $F(p) = \{x : x \notin \text{arms}(p) \text{ and } \text{opp}(x) \notin \text{arms}(p)\}$. With m_x people waiting to cross arm x , k_x currently on that crossing and v_x the longest pedestrian wait there,

$$\Phi(p) = w_{\text{ped}} \left(\sum_{x \in F(p)} (m_x + k_x) \right) \left(1 + \frac{\min(\max_{x \in F(p), m_x > 0} v_x, V)}{V} \right), \quad (5)$$

with $w_{\text{ped}} = 0.3$, so one waiting person contributes about as much pressure as one motorcycle, and V the pedestrian maximum-wait parameter. Turning traffic across an open crossing yields to pedestrians in normal permissive operation, which is why $F(p)$ excludes only straight-ahead conflicts; counting pedestrians whose crossing is not actually freed causes the controller to hold greens that do not serve them. The complete score adds a large emergency constant $B(p) = B_0 \mathbb{1}[\exists a \in p : a \in \hat{E}]$ with $B_0 = 10^6$:

$$S(p) = D(p) + \Phi(p) + B(p). \quad (6)$$

A detection is a per-frame event and detectors drop frames, so each approach carries a hold timer reset to $\theta = 2.5$ s whenever an emergency is announced on that arm and otherwise decaying by Δt ; the latched set is $\hat{E}$. Announcements arrive at arm level and every movement key of that arm is latched, because the camera cannot resolve which lane the emergency vehicle occupies, so a single missed detection cannot abandon an active preemption.

4.3 Online estimation of arrival and discharge

Both remaining rules need to know how fast an approach fills and how fast it empties. A fixed-time plan obtains the equivalent from a traffic survey and a saturation-flow table. We estimate both from the count sequence the detector already produces, which keeps the controller free of any per-site constant and lets it follow a junction whose behaviour changes with the hour and the weather.

Let $d_a(t)$ be the PCU demand on approach a at step t and Δt the step. An approach on red cannot discharge, so every increase in its queue is arrivals; an approach on green moves by arrivals less departures, so departures are the arrival rate less the observed change:

$$\hat{\lambda}_a \leftarrow \hat{\lambda}_a + \alpha \left(\frac{d_a(t) - d_a(t-1)}{\Delta t} - \hat{\lambda}_a \right) \quad (a \text{ on red}), \quad (7)$$

$$\hat{s}_a \leftarrow \hat{s}_a + \alpha \left(\hat{\lambda}_a - \frac{d_a(t) - d_a(t-1)}{\Delta t} - \hat{s}_a \right) \quad (a \text{ on green, } d_a(t-1) > 0), \quad (8)$$

with $\alpha = \Delta t / \tau$ and both estimates clamped below at zero. Two details are necessary rather than cosmetic. The clamp is applied to the running estimate and never to the individual sample: a per-step count moves in whole vehicles, so clamping samples discards only the low side of the noise and inflates $\hat{s}_a$, measured at 0.79 against a ground truth of 0.55 PCU/s. Equation 8 is conditioned on the queue held *before* the step and not on the one after it, because whether an approach ends a step non-empty depends on whether a vehicle happened to arrive during it, and conditioning on that selects the steps with arrivals and deflates the same estimate to 0.30. With both corrections the estimator recovers 0.51 to 0.58 PCU/s against a ground truth of 0.52 to 0.55 across demand levels, and $\hat{\lambda}_a$ to within roughly 20%.

Both estimates begin at zero, which makes the two rules below inert until the junction has been observed for a few cycles; the controller reduces to a plain hysteresis policy in the interim.

4.4 Green duration from the measured cycle

Webster [1] minimises delay at cycle length $C = (1.5L + 5)/(1 - Y)$, where L is the lost time per cycle and $Y = \sum_p y_p$ sums the critical flow ratios $y_p = \max_{a \in p} \lambda_a / s_a$, allocating each phase a green of $(C - L)y_p / Y$. The formula is normally applied offline to survey data. Equations 7 and 8 supply y_p directly, so the same allocation can be evaluated continuously from the junction in front of the camera:

$$g^*(p) = \max \left(g_{\min}, (C - L) \frac{y_p}{Y} \right), \quad y_p = \max_{a \in p} \frac{\hat{\lambda}_a}{\hat{s}_a}. \quad (9)$$

Y must sum the critical ratios of one cycle, so phases are accumulated heaviest-first over disjoint approach sets; on a movement-level junction the phase sets overlap, and summing all of them would invent saturation that is not present. Where $Y \geq 1$ no cycle length clears the demand and C does not exist, and the controller falls back to $g_{\max}$, the longest green its anti-hog bound permits, which is the throughput-maximising choice and the usual practical fallback.

$g^*(p)$ is evaluated once, at the instant the phase takes green, and acts as a floor on that green in place of $g_{\min}$ in rule 4 of the stage machine. Evaluating it continuously would chase a target that falls as the queue drains, ending the green for the reason it was extended. It is a floor and not a hold: emergency preemption and the empty-phase skip are both evaluated above it and

continue to use $g_{\min}$, so nothing waits behind a bound intended for a busy junction. Under light demand Y is small, Equation 9 evaluates below $g_{\min}$, and the rule never binds.

It is deliberately not capped at $g_{\max}$. That parameter bounds how long one movement may hold the junction, which is a fairness policy; refusing a green the measured demand requires does not protect the minor approaches, because the queue that was not cleared is still present at the next cycle.

4.5 Pricing a phase change

A phase change costs $y + \rho$ seconds during which no approach discharges. The approaches currently green would have spent that interval discharging, so the challenger must beat the incumbent by more than what the handover forfeits:

$$\kappa(p) = \sum_{a \in p} \min(d_a, \hat{s}_a(y + \rho)), \quad (10)$$

and rule 9 of the stage machine switches only when $S(p^*) > S(p) + h + \kappa(p)$. The minimum in Equation 10 caps the forfeit at what the approach still has queued, so a drained or empty phase costs nothing to leave and the gap-out and empty-phase rules retain their original sensitivity. A constant margin cannot express this: h is the same whether the junction is empty or saturated, whereas κ is zero on an empty junction and grows with the traffic actually mid-discharge.

4.6 Target selection and the stage machine

The signal cycles through green, yellow of duration y and all-red of duration ρ ; a green is never followed directly by a conflicting green. The next phase is chosen by strict precedence rather than by the score alone. *(i) Emergency:* if $\hat{E} \neq \emptyset$, take the phase serving the most emergency approaches, breaking ties by the largest r_a among them, so that on a movement-level junction the split-arm phase is chosen and whichever lane the emergency vehicle occupies turns green. *(ii) Vehicle starvation:* otherwise, if any approach has $r_a \geq R$ and $d_a > 0$, force a phase serving the worst such approach, preferring one that also serves approaches within 25 s of the threshold so that two clearance intervals are not paid where one would do; an empty approach cannot starve, so this never awards green to an empty lane. Candidates are restricted to approaches that are not currently green and whose r_a exceeds that of every approach the current phase is serving. Both restrictions carry the same meaning: this rule exists to serve the worst-treated queue, not to rotate. Beyond capacity every approach sits past R permanently, and a rule that fires whenever anyone is past the threshold fires on every step (Section 6.2). *(iii) Pedestrian starvation:* otherwise, if any arm has $m_x > 0$ and $v_x \geq V$, force a phase in $F(x)$ with the greatest pedestrian demand. *(iv) Score:* otherwise take $\arg \max_p S(p)$ over phases with $D(p) > 0$ or non-zero pedestrian demand, resting in all-red if no phase has demand.

A running green is terminated by the first applicable rule below, where e is elapsed green, $g_{\min}$ and $g_{\max}$ the minimum and maximum green, γ_t the gap-out time, Θ the preemption cap, W the walk-hold allowance and h the hysteresis margin.

1. An emergency exists on another phase and not the current one, and either $e \geq g_{\min}$ or the current phase is empty.
2. An emergency exists on the current phase: hold, subject to two bounds. An approach reaching $r_a \geq 2R$ forces service even over an active hold, and the cumulative hold is capped at Θ , after which the emergency term is dropped so the ordinary rules can move the signal. This prevents a stationary emergency vehicle from freezing the junction.
3. The current phase is empty, meaning no approach in p has $o_a = 1$ and no freed crossing has pedestrian demand, while another phase has demand: switch immediately without waiting for $g_{\min}$, since an empty phase has no vehicle to strand.

Table 2: Controller parameters. The single-junction column is the configuration used in Experiments 1 and 3; the corridor column is the configuration of the system’s network demonstration, used in Experiment 2.

Symbol	Meaning	Single junction	Corridor
$g_{\min}$	minimum green (s)	5.0	5.0
$g_{\max}$	maximum green (s)	30.0	34.0
τ	estimator time constant (s)	60.0	60.0
y	yellow (s)	3.0	2.0
ρ	all-red clearance (s)	1.5	2.5
γ_t	gap-out time (s)	2.5	2.5
R	vehicle maximum wait (s)	60.0	26.0
V	pedestrian maximum wait (s)	60.0	60.0
W	walk-hold allowance (s)	12.0	12.0
Θ	preemption cap (s)	60.0	25.0
θ	emergency latch (s)	2.5	2.5
h	hysteresis margin (PCU)	1.0	1.0
w_{ped}	pedestrian weight	0.3	0.3
γ	downstream coupling	n/a	0.4

4. $e < g^*(p)$: hold, where $g^*(p) \geq g_{\min}$ is the green allocated at onset by Equation 9. Minimum green is not overridden for a phase actually serving traffic, and under measured saturation the green worth serving is longer than the safety minimum.
5. Vehicle starvation elsewhere, then pedestrian starvation elsewhere: switch.
6. A pedestrian is still on a crossing freed by the current phase and $e < \min(g_{\max}, g_{\min} + W)$: hold. The bound matters, because an unbounded hold lets a renewing pedestrian stream pin green on a lightly used approach while a loaded corridor waits.
7. Gap-out: the served approaches have read empty for at least γ_t and another phase has demand.
8. Maximum green: $e \geq g_{\max}$, so green is yielded to the best other phase with demand.
9. Score, against the cost of switching: $S(p^*) > S(p) + h + \kappa(p)$ for the best alternative p^* , with $\kappa(p)$ from Equation 10. The constant term suppresses oscillation between phases of nearly equal score on an idle junction; the measured term makes the required margin scale with the discharge the handover would forfeit.

Parameter values are listed in Table 2.

4.7 Corridor coupling and the preemption bound

On a corridor each junction j knows, for some arms, the downstream approach (j', a') that the arm’s discharge feeds. The demand presented to the score is then discounted by the downstream queue $q_{j', a'}$,

$$\tilde{d}_a = \begin{cases} \max(d_a - \gamma q_{j', a'}, \varepsilon) & o_a = 1, \\ 0 & o_a = 0, \end{cases} \quad \gamma = 0.4, \varepsilon = 0.05. \quad (11)$$

The floor at ε keeps the coupling a discount on pressure magnitude rather than an erasure of presence, since a physically occupied approach must remain occupied for Equations (2) and (3). Equation (11) is a one-sided approximation of the pressure difference used in max-pressure theory, where the difference is taken over turning movements with movement proportions; we use the

aggregate downstream queue because it is the only downstream quantity one camera per junction can supply.

The worst-case delay from the first emergency detection to green on the emergency approach is bounded by the case in which the announcement arrives immediately after the controller has committed to a transition away from the very phase that serves the emergency. The committed transition completes, the newly served phase must be given its minimum green, and only then can the preemption take effect through a second clearance interval:

$$L_{\max} = y + \rho + g_{\min} + y + \rho. \quad (12)$$

With the single-junction parameters of Table 2, $L_{\max} = 14.0$ s.

The system also ships an executable self-check asserting twelve properties across four groups: clearance, green allocation (no green to an empty approach while another has demand, a full window for a lone motorcycle, and the $g_{\min}$ and $g_{\max}$ bounds), fairness (the R and V bounds and the walk hold), and emergency handling (prompt service, survival of a dropped detection frame mid-preemption, sequencing of emergencies on conflicting phases, and the Θ bound), plus input validation. All twelve pass on the version evaluated here and the captured output accompanies the experiment artefacts. These are executable assertions about behaviour under specified stimuli, not proofs, and they say nothing about the correctness of the perception input.

5 Experimental setup

5.1 Traffic model and baselines

Control results are obtained in a point-queue microsimulation, a standard abstraction in which delay accumulates at a single point rather than along a link [43]. While an approach displays green it discharges at $s = 0.55$ veh/s, a saturation headway of about 1.8 s, within the range capacity guidance gives for a through lane [44], accumulated as a fractional carry so a green shorter than one headway releases no vehicle, and with no banking of unused capacity while the queue is empty. Arrivals are Poisson per approach. We did not use an established microsimulator such as SUMO [45] because the controller under test is the system’s own code and the purpose is to drive that code on identical arrival streams; the cost is traffic-model fidelity, which we state as a limitation rather than defend.

The arrival process and the discharge model are the code shipped with the system under test, imported unchanged; the experiment scripts add warm-up handling, additional metrics and the baselines. Arrivals are drawn once per time step from a single seeded generator and pushed into every policy’s queue system, so the comparison is paired at the level of individual vehicles. Runs are 1800 s with the first 300 s discarded, because all queues start empty, and vehicles still queued at the end are counted with the wait accumulated so far, since discarding them would flatter whichever policy left the longer queue.

Two fixed-time baselines are used. The *equal-split* plan is a round-robin giving each phase 13.0 s of green, which with 3.0 s yellow and 1.5 s all-red produces a 35 s cycle; this is the baseline shipped with the system and represents the demand-blind plans the work aims to replace. The *Webster* plan takes cycle length and splits from Webster’s formulae [1] computed from the *true* mean demand of the run: with lost time $L = |P|(y + \rho)$ and critical flow ratios $y_i = \max_{a \in p_i} \lambda_a / s$ summing to Y , the cycle is $C = (1.5L + 5)/(1 - Y)$ clamped to $[30, 120]$ s and the effective green of phase i is $(C - L)y_i / Y$ floored at $g_{\min}$; when $Y \geq 0.95$ no finite Webster cycle exists and the cycle is set to the upper clamp with splits proportional to y_i . This baseline is deliberately generous to fixed-time control, being granted perfect knowledge of the average demand and re-derived for every demand cell, and it is included because the equal-split baseline alone makes an adaptive controller look better than it is.

5.2 Experiments

Experiment 1: single junction. A four-arm junction with two phases, NS and EW, under three demand profiles. *Asymmetric* sets per-approach rates of (0.150, 0.130, 0.040, 0.035) veh/s for N, S, E and W, the shape the shipped benchmark samples at random, fixed here so a demand level is a stated number; *balanced* sets all four approaches to 0.089 veh/s; *single* loads one approach at 0.250 veh/s with 0.035 veh/s on the others. Each profile is scaled by multipliers $m \in \{0.6, 0.8, 1.0, 1.2, 1.4\}$, giving total demands of roughly 770 to 1790 veh/h, and each cell is run with 30 seeds. We report mean, median, 95th-percentile and maximum wait per vehicle, mean and maximum queue length and throughput; per-seed reductions are compared with a paired t -test and a Wilcoxon signed-rank test. A companion diagnostic (Experiment 1b) measures, over 10 seeds per cell, the phase change rate and the fraction of wall time during which some approach displays green; it explains Experiment 1 rather than adding a claim.

Experiment 2: three-junction corridor. A straight arterial carrying a three-arm junction, a four-arm junction and a second three-arm junction at 90 m spacing, with the topology, phase sets, routes and timing constants of the system’s network demonstration, and six routes: two arterial through movements and four entering from side arms. Link travel time is $90/12.5 = 7.2$ s at a free-flow speed of 12.5 m/s, so free-flow traversal of the three links is 21.6 s. Four policies are compared: the controller with coupling (Equation (11)) active, the same controller with $\gamma = 0$, an independent Webster plan per junction, and a coordinated Webster plan using a common cycle equal to the longest Webster cycle on the corridor with an eastbound progression offset of one link travel time per junction. Runs are 2400 s with 400 s warm-up, 20 seeds, at three demand levels, and one emergency vehicle is released from the west at $t = 1200$ s in every run, with other traffic yielding to it so that its traversal time isolates the signal contribution. The traffic layer is a re-implementation, because the system’s corridor demonstration is a browser application that cannot be driven headlessly; the control logic is the system’s own code.

Experiment 3: preemption latency. The junction is driven to a loaded state for a random interval between 40 and 160 s, after which an emergency vehicle is announced on a randomly chosen approach and re-announced every step thereafter. We record the delay until that approach displays green, with the stage and elapsed green at the instant of announcement, and replay the identical trial against the equal-split baseline, which has no preemption path. 300 trials are run at each of three demand levels.

Experiment 4: perception. The shipped detector is evaluated on the system’s held-out split of 97 images containing 4990 labelled instances across five classes. The full perception stack is then run over 60 s of two real fixed-camera junction clips at 960 px inference on an Apple M3 Pro using the Metal backend, recording per-frame latency and per-approach counts, with the controller running on those counts for the clip that shows two competing signalised streams so that the signal state in the exported figures is its real output. Finally, to test whether the counts have converged, the stack is run over frames sampled every 2 s from each clip at 640, 960, 1280 and 1600 px with temporal smoothing disabled.

Experiment 6: robustness across demand shapes and topology. Experiment 1 reports means over three demand shapes, which can conceal the cells where an adaptive policy loses. This experiment widens the grid to seven demand shapes on the four-way junction, from perfectly balanced to a single dominant approach, plus two shapes on a three-arm T junction, each at eight load levels running past the point at which a fixed plan saturates, for 72 cells of 25 paired seeds. The reported quantity is per-cell rather than aggregate: how many cells lose to either baseline, and by how much in the worst one. The grid is then repeated with the counts presented to the controller corrupted by zero-mean Gaussian noise of 20% of each count, while the queues being managed remain exact, which separates the control policy’s robustness from the detector’s accuracy.

Table 3: Experiment 1. Mean delay per vehicle in seconds (standard deviation across 30 seeds in parentheses) and the paired per-seed reduction achieved by R.E.L.A.Y. against each baseline. Positive reductions favour R.E.L.A.Y.. All differences are significant at $p < 0.05$ by the Wilcoxon signed-rank test on paired per-seed means.

Profile	Demand (veh/h)	Mean delay per vehicle (s)			Reduction (%)	
		Equal split	Webster	R.E.L.A.Y.	vs equal	vs Webster
Asymmetric	767	11.0 (0.6)	8.4 (0.6)	5.0 (0.3)	+54.6 (3.8)	+40.1 (5.0)
	1022	11.8 (0.7)	8.9 (0.6)	5.5 (0.3)	+53.0 (4.1)	+37.7 (5.0)
	1278	13.9 (0.8)	9.6 (0.4)	6.3 (0.3)	+54.1 (4.5)	+33.4 (4.3)
	1534	19.2 (3.1)	10.7 (0.8)	7.2 (0.4)	+61.8 (6.2)	+32.7 (4.6)
	1789	45.1 (17.2)	11.8 (1.0)	8.1 (0.4)	+79.8 (6.8)	+31.4 (3.9)
Balanced	769	10.5 (0.6)	9.9 (0.4)	6.1 (0.3)	+42.0 (3.3)	+38.5 (3.0)
	1025	10.9 (0.4)	10.5 (0.4)	6.7 (0.3)	+38.0 (3.9)	+36.0 (3.4)
	1282	11.4 (0.3)	11.2 (0.4)	7.6 (0.3)	+33.8 (3.1)	+32.3 (2.9)
	1538	12.1 (0.6)	11.9 (0.6)	8.5 (0.4)	+29.9 (4.0)	+28.4 (3.7)
	1794	13.2 (0.6)	13.5 (0.6)	9.5 (0.3)	+27.9 (2.8)	+29.2 (3.1)
Single	767	14.4 (2.2)	8.8 (0.5)	5.4 (0.4)	+62.5 (4.0)	+39.3 (5.1)
	1022	38.3 (16.6)	9.8 (0.5)	6.5 (0.4)	+79.9 (8.2)	+34.0 (5.9)
	1278	151.8 (30.9)	11.8 (0.9)	7.7 (0.4)	+94.8 (1.0)	+34.9 (3.9)
	1534	239.8 (28.3)	17.1 (3.0)	9.3 (0.7)	+96.1 (0.5)	+44.3 (8.5)
	1789	307.2 (23.7)	21.3 (4.1)	14.0 (3.0)	+95.5 (0.8)	+33.0 (14.3)

6 Results

6.1 Single junction

Table 3 and Figure 1 give the single-junction results. R.E.L.A.Y. reduces mean delay against both baselines in all fifteen cells, by 28.4% to 44.3% against the Webster plan and by 27.9% to 96.1% against the equal-split plan.

Under asymmetric demand, the regime the system is designed for, R.E.L.A.Y. reduces mean delay by 54.1% against the equal-split plan and by 33.4% against the Webster plan at 1278 veh/h. The gain against the equal-split plan grows with demand, reaching 79.8% at 1789 veh/h, but that growth is a capacity artefact rather than a control improvement: at that demand the equal-split plan is oversaturated on the major approaches and its mean delay diverges, whereas the Webster plan and R.E.L.A.Y. both remain stable at essentially the same throughput, 1804 and 1801 veh/h. The gain against the Webster plan is close to flat across the range, 40.1% down to 31.4%. We regard the Webster column as the honest headline.

Under a single dominant approach the equal-split plan collapses: at 1278 veh/h its mean delay is 151.8 s, its mean queue is 55.4 vehicles and its throughput falls to 1123 veh/h against a demand of 1278 veh/h, so it is not serving the traffic offered. Against the Webster plan the gain peaks at 44.3% at 1534 veh/h, where the measured cycle is longest relative to the fixed plan’s, and falls to 33.0% at 1789 veh/h as both controllers approach the capacity of the loaded approach.

Under balanced demand, the hard case for a pressure policy and the case a fixed plan is designed for, the gain against the Webster plan is 38.5% at 769 veh/h and remains 29.2% at 1794 veh/h. This is the result that changed most between versions of this controller, and Section 6.2 reports what the earlier version did here and why.

Figure 2 shows the wait distribution rather than the mean, which is where the fairness bound becomes visible. At 1278 veh/h under asymmetric demand the 95th-percentile wait is 29.9 s under the equal-split plan, 25.8 s under the Webster plan and 14.5 s under R.E.L.A.Y., with worst observed waits averaging 50.0 s, 50.1 s and 26.7 s. Under a single dominant approach the contrast is larger: 333.4 s and 36.3 s at the 95th percentile against 17.0 s under R.E.L.A.Y., and worst

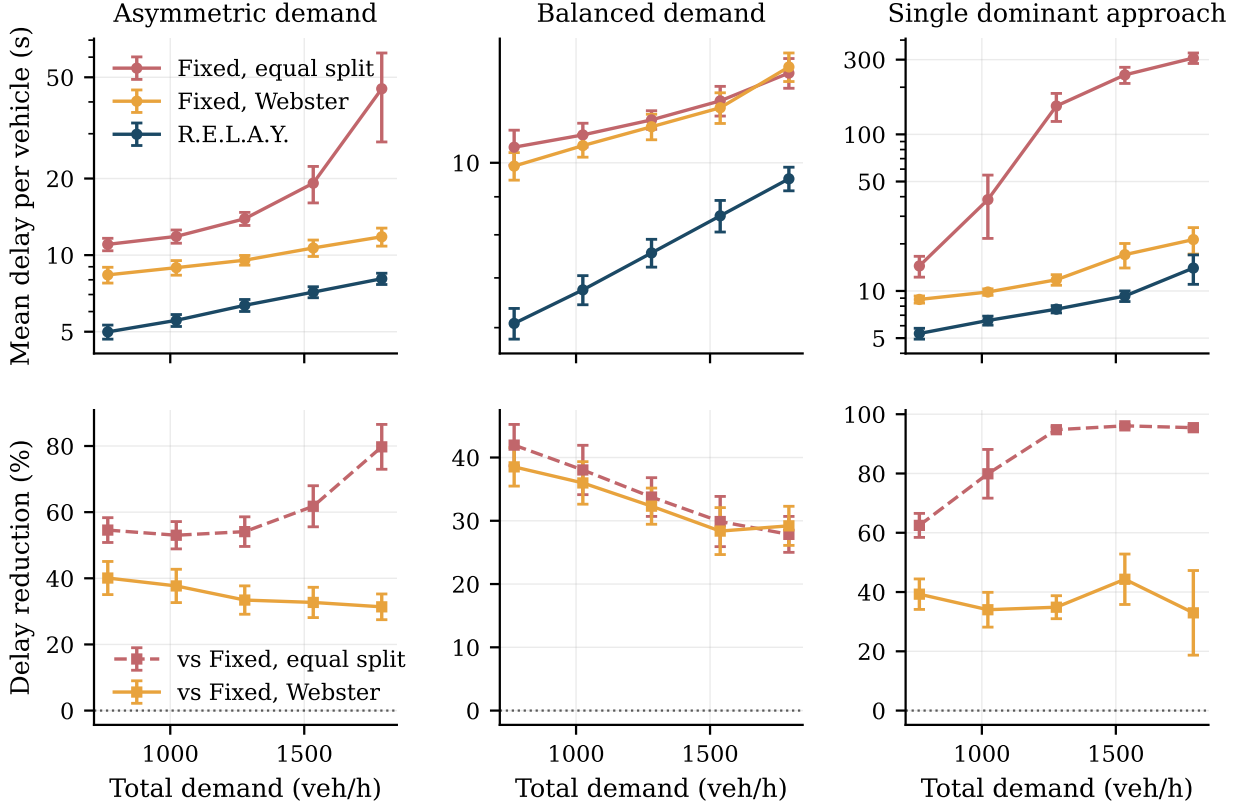


Figure 1: Experiment 1. Top row: mean delay per vehicle against total demand for the three demand profiles, on a logarithmic axis. Bottom row: paired per-seed delay reduction achieved by R.E.L.A.Y. against each fixed-time baseline. Error bars are one standard deviation across 30 seeds. Every cell lies above zero against both baselines.

waits of 361.5 s and 72.9 s against 36.9 s, consistent with the forced-service rule at $R = 60$ s. The tail improves in every cell, which matters more than the mean for a policy whose fairness claim is a bound rather than an average. Figure 3 shows the same comparison unaggregated, one point per seed.

6.2 Switching, clearance, and what saturation costs an unconstrained policy

Balanced saturated demand is where an unconstrained pressure policy fails, and the failure is worth stating precisely because it motivates three of the rules in Section 4. No phase dominates, so successive scores sit near-tied and any per-step comparison flips on noise. A constant hysteresis margin is crossed almost immediately. The anti-starvation rule, if it fires whenever any approach has waited past R , fires on every step, because beyond capacity every approach is past R permanently. Both terminate green at $g_{\min}$, and the junction is no longer being controlled, only alternated. Measured on this junction at 2307 veh/h, a policy with those two properties displays 5.2 s of green per 4.5 s of clearance interval, holds green for 0.54 of wall time against the fixed plan’s 0.80, and reaches 94 s of mean delay against that plan’s 19 s.

This is the practical form of a known theoretical issue: max-pressure as formulated in the literature evaluates its argmax freely and imposes no cycle structure, and Levin et al. [23] propose a cyclical structure to recover the lost capacity. Sections 4.4 and 4.5 recover it with a cycle measured rather than configured, and the starvation rule of Section 4 serves the worst-treated queue instead of rotating among equally starved ones. Each is separately necessary; none requires a site survey.

Figure 4 shows what the resulting policy does. R.E.L.A.Y. switches more often than the fixed plan under balanced demand, 282 to 331 per hour against 204, and displays green for less of the

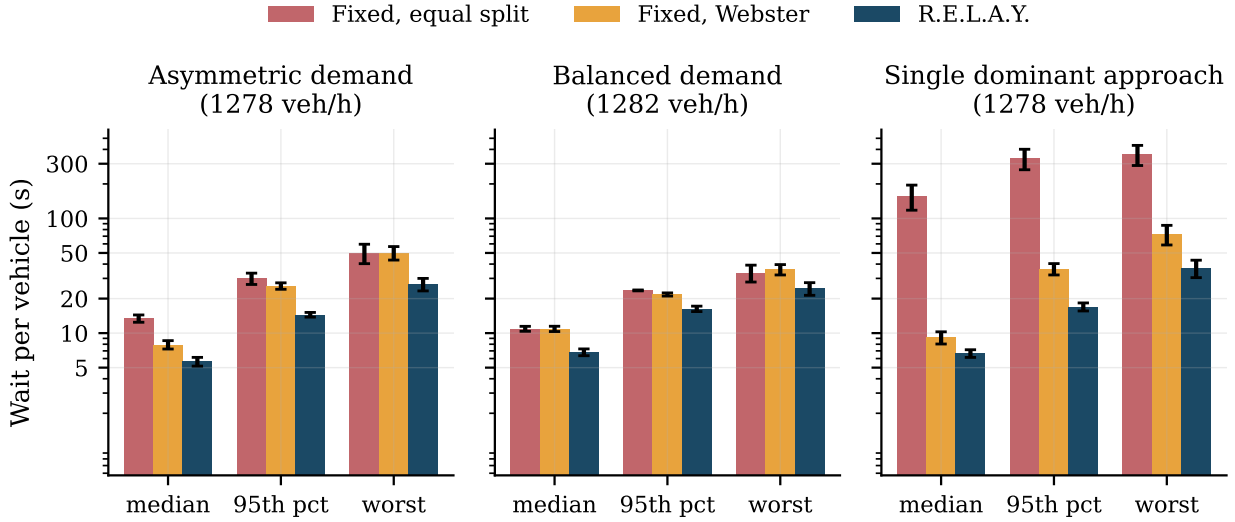


Figure 2: Experiment 1. Median, 95th-percentile and worst observed wait per vehicle at demand multiplier 1.0, on a logarithmic axis. The tail, not the mean, is where the anti-starvation bound shows: under a single dominant approach the equal-split plan’s worst wait averages 361.5 s against 36.9 s for R.E.L.A.Y..

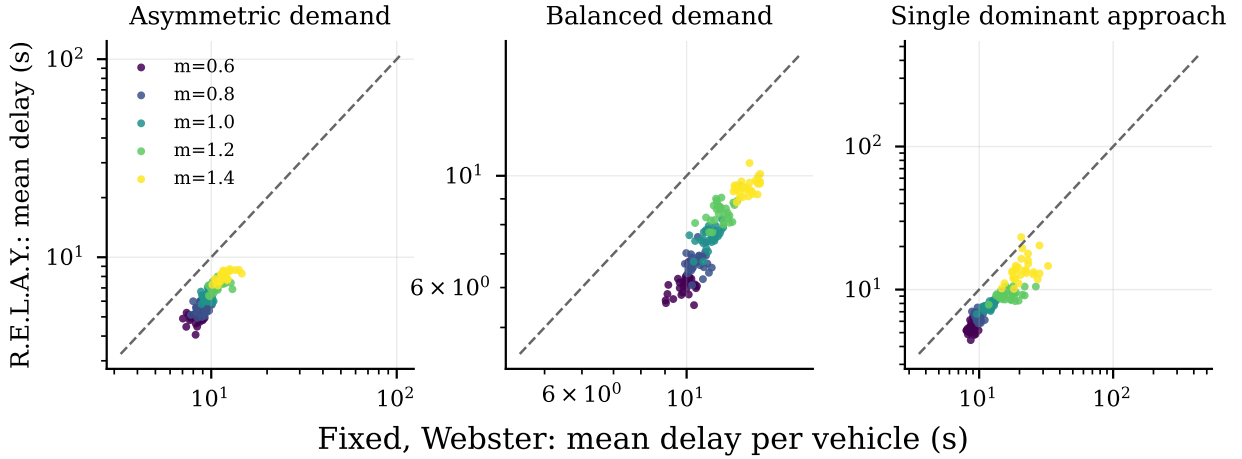


Figure 3: Experiment 1, unaggregated. Each point is one seed at one demand level, plotting R.E.L.A.Y.’s mean delay against the Webster plan’s mean delay on the identical arrival stream. Points below the diagonal favour R.E.L.A.Y.. Colour encodes the demand multiplier from 0.6 (dark) to 1.4 (light).

wall clock, 0.585 to 0.646 against 0.743, while reducing delay by 28.4% to 38.5%. Displayed green time is therefore not the objective: a fixed plan spends its green where the demand is not, and slightly less green in the right place beats more of it in the wrong one. Where both effects align the margin is largest, and under a single dominant approach at $m = 1.4$ R.E.L.A.Y. switches less than the fixed plan, 167 per hour against 204, and holds green longer, 0.791 against 0.743.

6.3 Corridor

Table 4 and Figure 5 contain three findings, one of them negative.

First, the delay reduction is large across the range and decays only gradually with demand: 57.9%, 49.4% and 44.6% against the independent Webster plans at 769, 1282 and 1794 veh/h, and 56.3%, 47.8% and 42.2% against the coordinated plan. The coordinated plan is only 2 to 4% better than the independent plans on mean delay, as expected, since progression offsets improve the through movement and not the average over all movements. Arterial through travel time and stops per vehicle improve together with delay: at the highest demand, 32.4 s and 1.86 stops

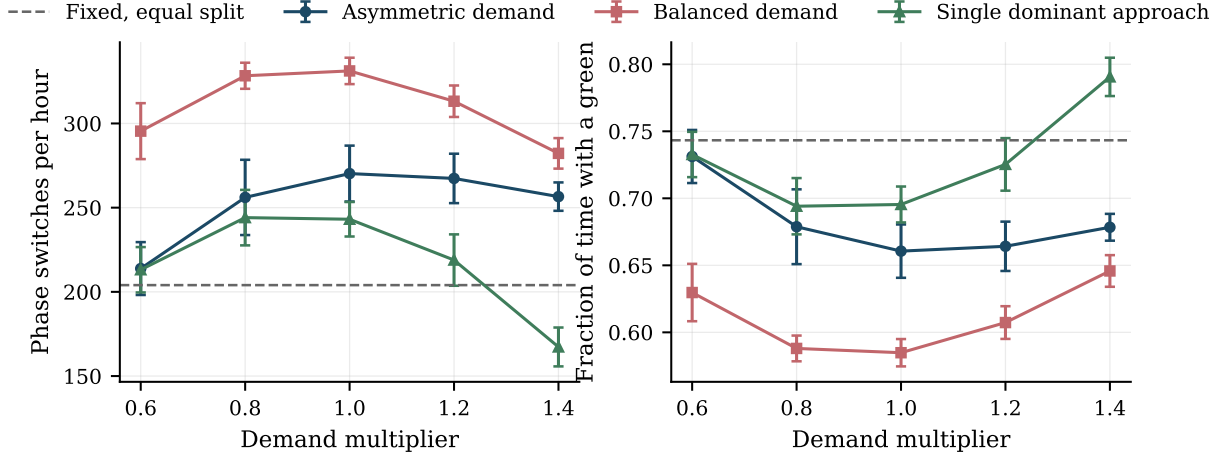


Figure 4: Experiment 1b. Left: phase changes per hour. Right: the fraction of wall time during which a green is displayed, the complement of which is time spent in yellow and all-red clearance. The dashed line is the equal-split fixed plan, whose behaviour is independent of demand by construction.

against the coordinated plan’s 39.3 s and 2.03, so the gain is not obtained by trading progression for total delay.

Second, the coupling term does essentially nothing. Disabling it, which turns the corridor into three independently adaptive junctions, changes mean delay by +0.1%, −0.5% and −0.9% at the three demand levels, and at the two higher levels the uncoupled variant is marginally better. The corridor benefit is therefore attributable to local adaptivity, not to coordination. We report this plainly because the coupling term is part of the system’s design rationale and the data do not support it. Two explanations are consistent with our setup and we cannot separate them here: the point-queue model has no spatial extent, so a downstream queue never physically blocks an upstream discharge, which removes the main mechanism the coupling addresses; and a single aggregate downstream queue is a weak proxy for the movement-specific pressure difference max-pressure theory uses. Distinguishing them requires a model with link storage, which we regard as the most informative single extension to this evaluation.

Third, the emergency result is unambiguous. The emergency vehicle traverses the corridor in 22.3 to 27.6 s under R.E.L.A.Y. against a free-flow reference of 21.6 s, so it is delayed by roughly 1 to 6 s across three signalised junctions, against 30.1 to 53.6 s under the independent Webster plans. The coordinated plan does well here, 24.0 to 31.5 s, because the emergency vehicle travels in the progression direction; a westbound emergency would not benefit, whereas R.E.L.A.Y. is direction-agnostic by construction.

6.4 Emergency preemption latency

Mean latency is 4.09 s against 7.00 s for the fixed cycle, and is almost insensitive to demand, rising only from 4.01 s to 4.19 s between the lightest and heaviest loads (Table 5). That insensitivity is a design property rather than a coincidence: the green allocated by Equation 9 grows with saturation, but preemption is evaluated above it against $g_{\min}$, so a busier junction does not make an emergency vehicle wait longer. The important quantity is the tail: the 95th percentile is 11.33 s against 20.59 s and the worst case over 900 trials is 13.99 s against 22.29 s. The measured worst case sits 0.01 s below the analytical bound of 14.0 s from Equation (12), and inspecting the worst trials confirms the predicted mechanism, since every trial above 13.4 s is one in which the announcement arrived within the first second of a yellow already transitioning away from the emergency’s own phase. Because the bound is composed only of clearance and minimum-green settings, it can be tightened by configuration and stated in advance to an emergency service, which a demand-blind fixed plan cannot do.

Table 4: Experiment 2. Three-junction corridor, mean over 20 seeds with standard deviation in parentheses. Delay is total queueing delay per vehicle across all junctions traversed. Through travel time is measured for the two arterial through routes only; free-flow traversal of the three links is 21.6 s. Emergency traversal is the single emergency vehicle released in each run, whose standard deviation is zero for the fixed plans because those plans are deterministic.

Demand (veh/h)	Policy	Mean delay per vehicle (s)	Through travel time (s)	Stops per vehicle	Emergency traversal (s)
769	Webster, independent	26.7 (0.9)	39.7 (0.9)	1.95	36.0 (0.0)
	Webster, coordinated	25.7 (0.7)	37.0 (0.7)	1.78	24.5 (0.0)
	R.E.L.A.Y., no coupling	11.3 (0.5)	24.9 (0.5)	1.09	22.3 (3.3)
	R.E.L.A.Y., coupled	11.2 (0.5)	24.9 (0.5)	1.09	22.3 (3.3)
1282	Webster, independent	30.1 (0.8)	41.5 (0.7)	2.09	53.6 (7.5)
	Webster, coordinated	29.2 (1.1)	38.0 (0.8)	1.80	31.5 (0.0)
	R.E.L.A.Y., no coupling	15.2 (0.7)	28.3 (0.6)	1.53	23.7 (4.7)
	R.E.L.A.Y., coupled	15.2 (0.6)	28.5 (0.6)	1.53	24.2 (6.7)
1794	Webster, independent	36.7 (3.3)	43.1 (1.3)	2.10	30.1 (6.5)
	Webster, coordinated	35.3 (3.6)	39.3 (0.9)	2.03	24.0 (0.0)
	R.E.L.A.Y., no coupling	20.0 (0.8)	31.9 (0.5)	1.86	26.1 (8.5)
	R.E.L.A.Y., coupled	20.2 (0.8)	32.4 (0.7)	1.86	27.6 (8.2)

Table 5: Experiment 3. Delay from the first emergency detection to green on the emergency approach, over 900 trials. The equal-split baseline has no preemption path, so its latency is determined by cycle position and is independent of demand, which is why its statistics are pooled.

Demand	Policy	Mean (s)	SD (s)	Median (s)	95th pct (s)	Worst (s)
Light ($m=0.6$)	R.E.L.A.Y.	4.01	3.55	4.58	10.77	13.61
Medium ($m=1.0$)	R.E.L.A.Y.	4.06	3.90	4.31	11.88	13.91
Heavy ($m=1.4$)	R.E.L.A.Y.	4.19	3.89	4.54	11.85	13.99
All	R.E.L.A.Y.	4.09	3.78	4.54	11.33	13.99
All	Equal split	7.00	7.04	5.02	20.59	22.29

Two qualifications belong with this result. In 311 of 900 trials, or 34.6%, the emergency approach already had green when the vehicle was detected, so latency was effectively zero and the mean is pulled down accordingly; and in 298 trials, or 33.1%, the fixed cycle happened to reach that approach sooner than the preemption did, which is why the distributions overlap substantially in Figure 6 even though tail behaviour differs sharply. The claim these data support is a bounded worst case, not a uniformly faster response.

6.5 Perception

Table 6 reports mean average precision of 0.954 at IoU 0.5 and 0.809 averaged over IoU 0.5 to 0.95, with precision 0.950 and recall 0.886. Per-class average precision exceeds 0.96 for cars, buses, trucks and ambulances and is lowest for motorcycles at 0.851, the expected ordering given that two-wheelers are the smallest and most frequently occluded class. An important caveat concerns what the split is: it mixes automatically labelled frames from a synthetic feed with frames from real footage pseudo-labelled by a general detector rather than by human annotation, so a high score establishes that the fine-tuned model reproduces the labelling function it was trained against, not accuracy against human ground truth, and the ambulance figure of 0.987 reflects a class that is easy to separate in the synthetic domain. We deliberately omit a comparison against general COCO-trained checkpoints on this split, because their class indices do not correspond to this label space and any such comparison would be an artefact of index mismatch.

Running the full perception stack on real footage at 960 px inference on an Apple M3 Pro,

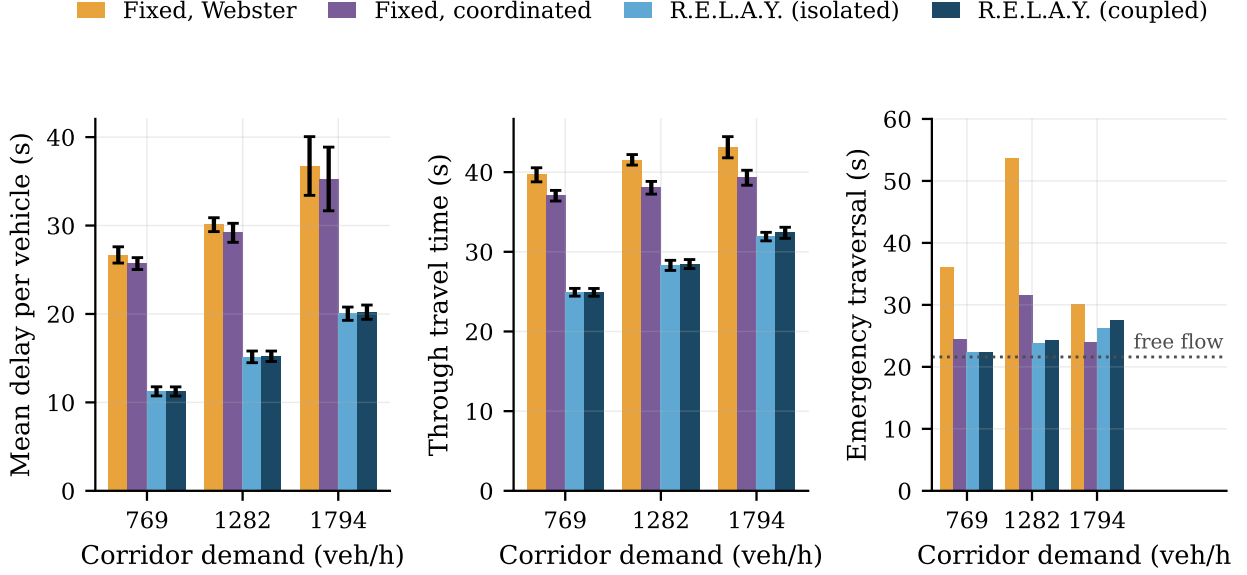


Figure 5: Experiment 2. Left: mean queueing delay per vehicle across the corridor. Centre: travel time for the two arterial through routes, where both adaptive variants stay below the coordinated fixed plan at every demand level. Right: traversal time of the single emergency vehicle released in each run, against the 21.6 s free-flow reference.

Table 6: Experiment 4. Detector performance on the system’s held-out split: 97 images, 4990 labelled instances. The checkpoint evaluated is the one shipped with the system, a 2.59-million parameter model. P and R are precision and recall at the operating point the validator reports.

Model	Overall				AP@50 by class				
	mAP@50	mAP@50-95	P	R	car	motorcycle	bus	truck	ambulance
Shipped fine-tune	0.954	0.809	0.950	0.886	0.985	0.851	0.981	0.964	0.987

median per-frame latency is 38.3 ms on the boulevard clip (26.1 frames per second, 95th percentile 85.7 ms) and 85.8 ms on the dense mixed-traffic clip (11.7 frames per second, 95th percentile 130.7 ms); the first frame of a run costs 1735 ms because of one-off backend initialisation and is excluded. Both rates exceed what the control loop needs, since decision epochs are governed by the 5 s minimum green rather than by the frame rate, and perception is decoupled from control so a latency spike delays a count update rather than blocking a decision.

Figure 8 shows annotated frames from both clips, and the second shows the main perception weakness plainly. To quantify rather than assert it, we measured how the count changes with inference resolution (Figure 7, right). On the dense clip the mean sampled vehicle count rises from 21.4 at 640 px to 26.5 at 960 px, 33.2 at 1280 px and 39.1 at 1600 px, an increase of 83% with no sign of convergence, and motorcycles alone rise by 120%; on the sparser boulevard clip the count rises from 4.85 to 10.0. A count that keeps growing with resolution has not converged, so the absolute demand values the controller receives on dense two-wheeler scenes are underestimates of unknown magnitude. Median inference cost over the same range rises from 47.2 ms to 108.0 ms, so raising resolution is not free.

This weakness does not immediately invalidate the control results, because the controller compares approaches rather than reading absolute volumes, so a systematic undercount affecting both approaches similarly leaves the ordering of phase scores intact. What it does affect is any rule keyed to an absolute quantity, in particular the occupancy indicator o_a near an empty approach and the gap-out timer. A regionally fine-tuned detector on human-annotated South Asian footage is the direct remedy and is not attempted here.

Taken together, these results support a narrower claim than the one usually made for camera-

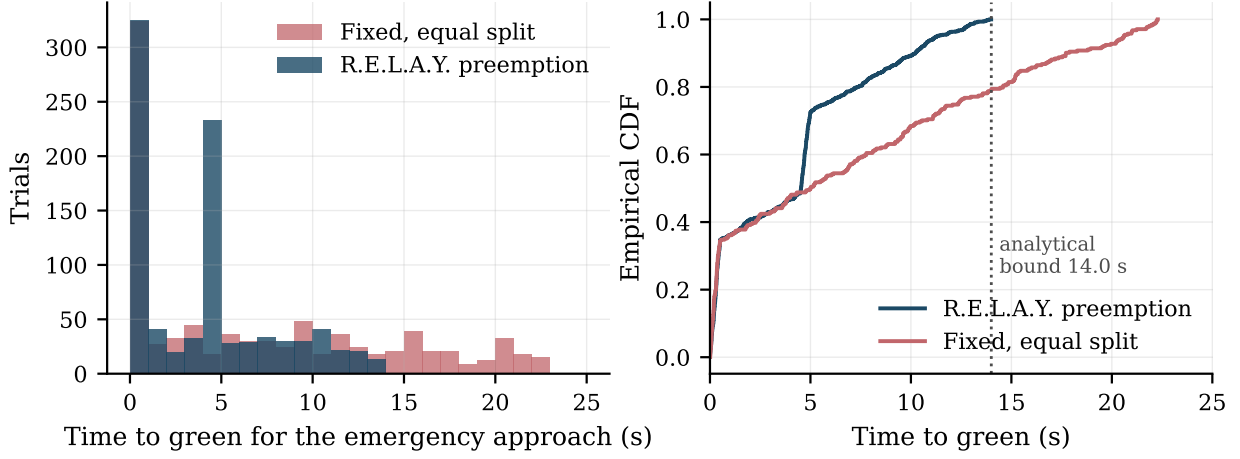


Figure 6: Experiment 3. Left: histogram of the time from the first emergency detection to green on the emergency approach, over 900 trials. Right: the same data as empirical cumulative distributions against the analytical bound of Equation (12).

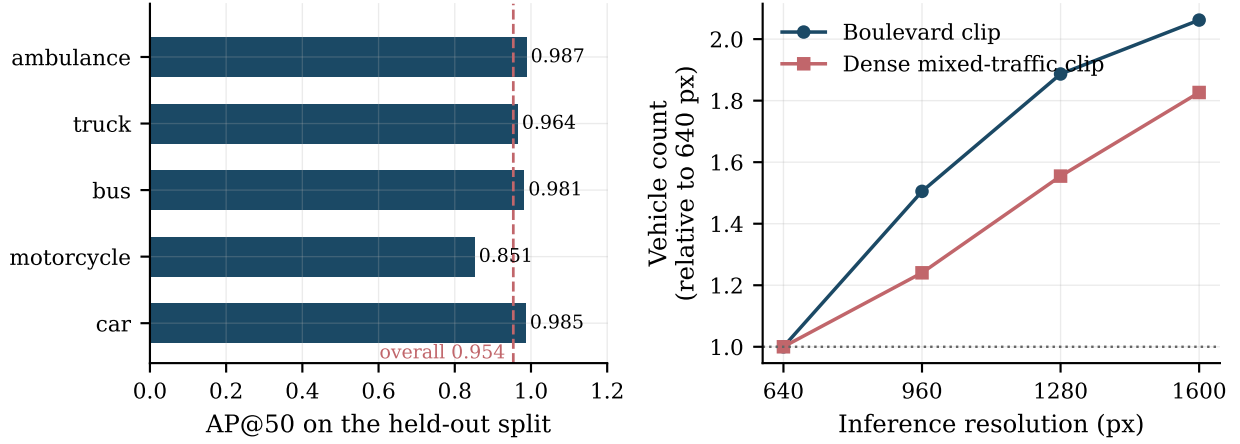


Figure 7: Experiment 4. Left: per-class average precision at IoU 0.5 on the held-out split, with the overall mean average precision marked. Right: sampled vehicle count as a function of inference resolution, normalised to the count at 640 px. Neither curve has flattened by 1600 px, which is direct evidence that vehicles are still being missed.

based adaptive control. The controller helps most where demand is imbalanced and the junction is not saturated, and its operational value is clearest in the bounds rather than the means: the worst observed wait tracks the configured R across every profile and the preemption worst case is bounded in closed form, so an authority can be told in advance what the guarantees are. The coordination term is not supported by our data, and the pedestrian logic, though verified by executable invariants, was never benchmarked against a pedestrian-aware baseline, so we claim no pedestrian delay benefit.

6.6 Robustness across demand shapes and topology

Table 7 reports the wider grid. R.E.L.A.Y. reduces mean delay against both fixed-time baselines in 71 of 72 cells, spanning 432 to 2462 veh/h, and never falls below +26.1% against the equal-split plan anywhere in the grid.

The single exception is the single-dominant-approach shape at the highest load, where the dominant approach alone demands 1620 veh/h against a saturation flow of 1980 veh/h, or 82% of all displayed green. Balancing that against the 9 s of lost time a two-stage cycle costs requires a cycle of at least 134 s, which is longer than the 60 s fairness bound R configured for these runs

Table 7: Experiment 6. Worst and median paired delay reduction over the eight load levels of each demand shape, 25 seeds per cell. Positive favours R.E.L.A.Y.. The worst column is the reportable one: an adaptive policy that wins on average and loses at one shape is not deployable.

Topology	Demand shape	Worst vs equal (%)	Worst vs Webster (%)	Median vs Webster (%)
Four-way	Asymmetric	+53.3	+29.2	+34.4
	Balanced	+27.1	+27.0	+32.5
	Near-equal	+28.5	+27.9	+33.0
	One axis	+63.4	+29.3	+38.8
	Three heavy	+33.1	+34.3	+36.0
	Lopsided	+52.6	+30.5	+34.8
	Single	+61.3	-26.6	+38.5
T-junction	Balanced	+37.1	+37.0	+39.6
	Arterial	+63.1	+30.5	+40.6

permits. The Webster reference is not subject to that constraint: it is granted a 120 s cycle and holds the minor approach on red for 107 s, and it is itself unstable there, with a residual queue at the end of the measurement window. Throughput is close to parity, 2222 against 2238 veh/h, and R.E.L.A.Y.’s mean delay is 26.6% higher. The honest description is that neither controller serves this demand, and that the comparison rewards the one that abandons the fairness bound. An operator who wants the Webster allocation here can have it by raising R ; that is a policy decision about how long a minor approach may be held, not a control-law result, and we do not report it as a win.

Repeating the entire grid with 20% zero-mean Gaussian miscounting on every count the controller sees leaves the picture unchanged: 71 of 72 cells still win, the worst cell against the equal-split plan moves from +27.1% to +26.1%, and the same single cell is the exception. The estimators of Section 4.3 smooth over a $\tau = 60$ s window, which is long relative to the noise, and the decisions they feed are floors and margins rather than instantaneous set points.

That test is the easy one. A zero-mean error averages out over the smoothing window by construction, whereas Section 6.5 measures a *biased* one: the detector reports 26 vehicles in a scene holding several hundred, and counts still climb 83% between 640 px and 1600 px without converging. We therefore repeat the grid under two count models that are wrong in the direction the detector is wrong. Queues stay exact in both, so what is measured is the control response to a wrong observation.

The first saturates, $\hat{q} = q/(1 + q/C)$: near-exact on a light approach, increasingly short as it fills, which is the shape of the measured failure. At $C = 15$ and $C = 30$ the result is unchanged, 71 of 72 cells, worst margin +26.6% against the equal-split plan, and a mean cost across the grid of -0.3 percentage points. The largest single-cell loss is 1.6 points. Saturation preserves the ordering of approaches and stays accurate where the estimators work, distorting only a tail that was going to be served anyway. The one cell the controller loses improves under it, from -26.6% to -4.2%, because compressing the dominant approach’s count stops the policy contending against the fairness bound.

The second is a flat multiplicative loss, $\hat{q} = \lfloor \kappa q \rfloor$. Down to $\kappa = 0.6$ it costs nothing measurable: 71 of 72 cells at $\kappa = 0.9, 0.8, 0.7$ and 0.6 , mean cost at most 0.1 points. That is a property of the formulation, not a lucky range. The degree of saturation in Section 4.4 is a ratio of an arrival estimate to a discharge estimate drawn from the same counts, so a uniform scale factor cancels before it reaches a decision.

At $\kappa = 0.5$ the controller collapses to 50 of 72 cells and a mean cost of 32.7 points, behind even the equal-split plan in the worst cell (-11.1%). The discontinuity is not a magnitude effect. At $\kappa = 0.5$ a lone vehicle rounds to zero, an occupied approach reports empty, and empty-phase skipping declines to serve it; at $\kappa = 0.6$ it survives rounding. Repeating $\kappa = 0.5$ with the single

change that an occupied approach may never report empty restores the result completely: 71 of 72 cells, worst margin +26.4%, mean cost 0.2 points. The 50% magnitude error was never the problem.

The requirement this yields is narrower and more testable than "the detector must be accurate". What the controller needs is an unbiased *presence* signal at short queue lengths, not an accurate absolute count. A detector that undercounts a crowd by an order of magnitude is tolerable, because both the pressure ordering and the saturation ratio survive it; one that cannot separate an empty approach from a nearly empty one is not, whatever its aggregate accuracy. That is a condition a field trial can check on a candidate detector before a signal is connected to it, and it is one the stack of Section 6.5 has not been shown to meet.

7 Limitations

We list the limitations we believe would most affect a reader’s assessment, in descending order of importance.

This is not a field study. No signal was controlled. There is no controller cabinet in the loop, no communication with a real signal and no before-and-after measurement, so we have no evidence about the failure modes that appear only in the field: power interruption, lens contamination, camera repositioning, seasonal light changes, or the behaviour of drivers who learn that the signal responds to them. Every quantitative claim should be read with that scope.

The traffic model is a point queue. Queues have no spatial extent, so there is no spillback, no blocking of an upstream junction by a downstream queue, no lane changing and no capacity loss from vehicles obstructing the junction box. The abstraction is favourable to any adaptive policy, whose mistakes are cheaper when queues cannot physically block anything, and specifically unfavourable to our coordination term, whose purpose is to prevent a jam being fed, so the corridor coupling result should be read as inconclusive rather than negative and a spatially explicit model in SUMO [45] would settle it.

Perception was neither validated against human ground truth nor propagated into the control experiments. The held-out split mixes synthetic frames with pseudo-labelled real frames, so the reported mean average precision of 0.954 measures agreement with the labelling process rather than accuracy against human annotation; we can state that counts on dense footage are underestimates that have not converged by 1600 px inference, but not the error magnitude, and a human-annotated evaluation set from the target city is required before any absolute demand claim. Experiment 6 addresses only part of the consequence: it shows the control policy tolerates 20% zero-mean miscounting, but real detector error is neither zero-mean nor independent across approaches, and undercounting two-wheelers biases whole phases in one direction, which a symmetric noise model cannot reproduce. Measuring how much of the delay reduction survives the detector’s actual, structured error remains the most important open question for the design.

The PCU weights are configuration, not calibration. The values in Table 1 were chosen to reflect a motorcycle-dominated mix rather than estimated from local field measurements. Because they directly scale the pressure in Equation (4), a mis-specified motorcycle weight biases every phase comparison at a junction whose approaches differ in class mixture, and we ran no sensitivity analysis over them.

The loop is never closed. Control results come from simulated counts and perception results from recorded footage, and at no point in this work does the detector drive the controller. The two halves are evaluated against each other only through the count-error models of Section 6.6, which is an argument about robustness rather than a demonstration of the assembled system. Those models are chosen to match the error the detector measurably makes and the controller survives them, but a genuine end-to-end run, detector to signal head on the same footage, is the obvious next experiment and it is not in this paper.

Four narrower gaps. The evaluated junctions have two stages, so although the controller generates movement-level phasing and protected-left stages, the results do not exercise them; a junction with five or six stages carries proportionally more lost time per cycle, and while Equation 9 lengthens the cycle in response, we have not measured whether it does so correctly when the disjoint cover has more than two members. We compare only against fixed-time plans, whereas the natural research baseline for a pressure-based controller is a learned policy, PressLight [29] in particular since its reward derives from max-pressure; our reasons for not deploying one are practical and stated in Section 2, but they do not excuse the absence of a comparison that would locate this controller between fixed-time and learned control. The pedestrian logic is asserted by executable invariants but never compared against an alternative treatment such as a conventional fixed pedestrian stage, so we report its behaviour and not its benefit. Finally, the fairness bound R is the parameter the results are most sensitive to: Section 6.6 shows it, and not the pressure rule, decides the one cell where a fixed plan wins, and we have not characterised the delay-fairness frontier it traces.

8 Conclusion

We have described a single-camera adaptive signal controller for heterogeneous urban traffic and evaluated it in a seeded microsimulation and on real fixed-camera footage. The controller converts per-approach, per-class detections into PCU-weighted demand and selects phases by a queue-weighted pressure score in the max-pressure family, with the operational rules a deployable signal requires: clearance intervals, minimum and maximum green, empty-phase skipping, gap-out, a hard maximum-wait bound for vehicles and a separate one for pedestrians, and emergency preemption with a bounded hold.

Its distinguishing feature is that it measures the junction rather than being told about it. Arrival and discharge rates are estimated per approach from the same count sequence the detector already produces, and they determine both the green each stage receives, through Webster’s cycle evaluated on the measured degree of saturation, and the margin a competing stage must clear, through the discharge that a clearance interval forfeits. Both are what a plain argmax policy lacks: without them, an unconstrained pressure rule terminates every green at the safety minimum once demand approaches capacity, spending the junction’s capacity on amber. Neither requires a traffic survey, a saturation-flow table, or any per-site constant, which matters for a deployment context where those are exactly what is unavailable.

Measured against a fixed-time plan granted the true mean demand, the controller reduces mean delay in every cell of both the primary grid and a wider 72-cell robustness grid but one, by 28.4% to 44.3% at a single junction and 44.6% to 57.9% across a three-junction corridor, cuts the 95th-percentile and worst-case waits proportionally further than it cuts the mean, and brings an emergency vehicle to green within a worst case of 13.99 s against an analytical bound of 14.0 s. The single exception is an approach demanding 82% of all displayed green, where the configured fairness bound forbids a cycle long enough to be stable and the fixed plan wins by not observing one.

One result is negative and reported rather than suppressed: the corridor coordination term produces no measurable benefit in our model, changing mean delay by at most 0.9%, so the corridor gain is attributable to local adaptivity alone. The next steps follow from the limitations: a spatially explicit corridor model to settle whether the coordination term is useful, a sensitivity study using the detector’s own structured error rather than symmetric noise, a human-annotated evaluation set from the target city, and a field trial.

Data and code availability

The controller, perception stack and simulation harness are available at <https://github.com/kafle1/relay> under the AGPL-3.0 licence. The experiment scripts that produced every table and figure, the raw per-run result files, the captured invariant-check output and the figure generation script are distributed with this manuscript in the `experiments/` directory. Re-running the experiment scripts and then the figure script reproduces every number and every panel reported here. The reproduction is exact rather than statistical: every run is seeded, and re-running Experiment 6 from a clean clone returns all 72 cells identical to the values reported here. A checker, `experiments/verify_claims.py`, reads the numerical claims out of this manuscript and recomputes each one from the raw result files, so a figure quoted in the text and the data behind it cannot drift apart unnoticed.

Declarations

The author declares no competing interests and received no funding for this work. No human subjects or personal data were involved: the footage used is publicly available recorded video of public roads, used only for aggregate vehicle counting, and no identification of individuals or vehicles was performed or attempted.

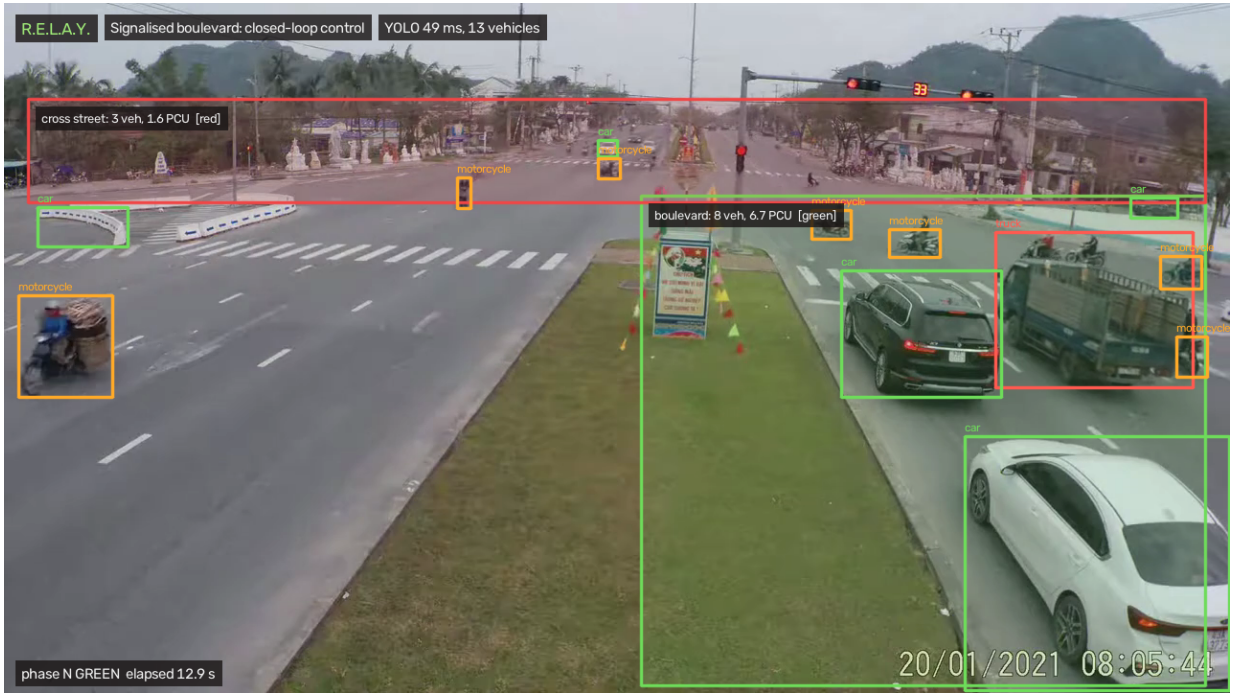
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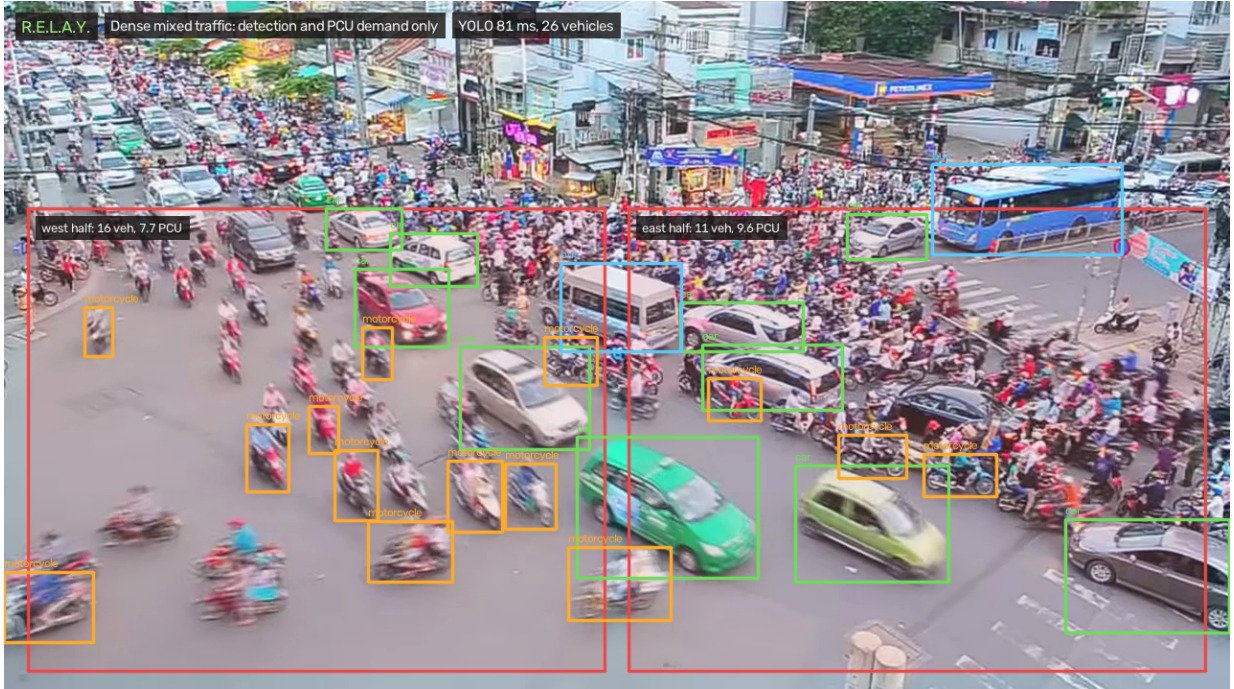
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(a) Signalised boulevard. The controller is running on these counts: the near boulevard approach holds 8 vehicles and 6.7 PCU and has green, the crossing street holds 3 vehicles and 1.6 PCU and is red, and the header reports the real inference latency for the frame.



(b) Dense motorcycle-dominated traffic. The failure mode is visible: 26 vehicles are detected in a scene containing several hundred, so the reported PCU demand of 7.7 and 9.6 for the two halves is a severe underestimate. No controller decision is shown, because the junction depicted is not signal-controlled in the way the two zones imply.

Figure 8: Experiment 4. The perception stack on real fixed-camera footage. Both frames are direct output of the system, with detections, approach polygons and per-approach PCU demand as computed at run time.